Cybersecurity Internship

WEEK 9 ASIGNMENT REPORT

Snort IDS – Custom Rules & Traffic Detection

Organization: DG Interns Hub

Submitted By: Amaan Fayaz

Project: Intrusion Detection System (IDS) Testing

Tools Used: Ubuntu Server, VMware Fusion, Snort IDS

Submission Date:

5TH NOV 2026

TASK 1 – SOC BASICS

SOC stands for Security Operations Center

A Security Operations Center is a team or department responsible for monitoring, detecting, analysing and responding to cybersecurity threats.

Its main purpose is to protect an organisation's computers, servers, networks, and data from cyberattacks.

Example: Suppose someone tries to hack a company's website. The SOC team monitors the network, detects suspicious activity, and investigates the attack.

Main responsibilities of SOC

1. Monitor network activities.
2. Detect suspicious activities.
3. Investigate security alerts.
4. Identify malware and cyberattacks.
5. Respond to security incidents.
6. Protect sensitive information.

2. Roles of SOC Analysts

There are three main levels of SOC analysts.

L1 – Level 1 Analyst (Security Monitoring)

- Monitors security alerts.
- Identifies suspicious activities.
- Check basic security incidents.
- Escalates serious issues to L2.

L2 – Level 2 Analyst (Incident Investigation)

- Investigate security incidents.
- Analyses suspicious activities.
- Identifies the source of attacks.
- Performs deeper security analysis.

- Escalates complex incidents to L3.

L3 – Level 3 Analyst Advanced Security Expert

- Handles complex cyberattacks.
- Performs advanced threat analysis.
- Conducts malware investigation.
- Develops security detection rules.
- Helps improve security systems.

Comparison Table

Basics	L1 analyst	L2 analyst	L3 analyst
Level	Beginner	Intermediate	Advanced
Main Role	Monitoring	Investigation	Advanced analysis
Responsibilities	Detect threats	Analyse incidents	Handle complex attacks
Alert Handling	Initial checking	Detailed investigation	Complex threat investigation
Tools	SIEM dashboards	SIEM, Wireshark, EDR	Advanced forensic and threat-hunting tools
Decision Making	Escalates incidents	Investigates and contains	Leads complex incident response
Experience	Entry-level	Experienced	Highly experienced

3. Difference Between SIEM, IDS and IPS

A. SIEM: Security Information and Event Management.

SIEM collects and analyses security logs from different devices and systems.

Purpose: Collect security information and identify suspicious activities.

B. IDS: Intrusion Detection System.

IDS monitors network traffic and detects suspicious activities. It generates alerts when it identifies potential threats.

Purpose: Detect threats and generate alerts.

C. IPS: Intrusion Prevention System.

IPS monitors network traffic and can automatically block or prevent detected malicious activities.

Purpose: Detect and prevent cyberattacks.

4. How SOC Works

A basic SOC follows this process:

1. Monitoring: Monitor network and system activities.
2. Detection: Identify suspicious activities.
3. Analysis: Investigate security alerts.
4. Response: Take action against threats.
5. Recovery: Restore affected systems and improve security.

Task 2 – Environment Setup.

1. Objective: The objective of this task was to set up a Linux-based environment for performing Security Operations Centre (SOC) activities and network security monitoring.

2. Virtual Machine Installation: I installed VMware Fusion on my MacBook Air M2 to create and manage a virtual machine. I configured the virtual machine with the required memory, processor, and storage.

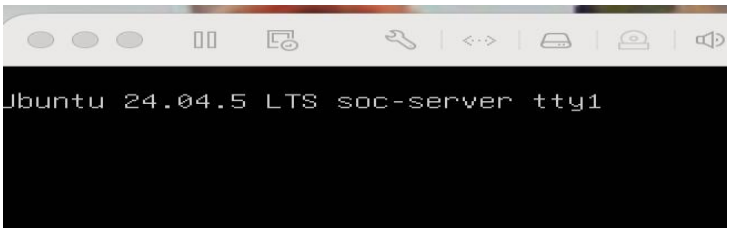


Figure 1: Ubuntu Server running inside VMware Fusion.

3. Ubuntu Server Installation: I downloaded and installed Ubuntu Server 24.04.5 LTS (ARM64) inside VMware Fusion. After installation, I successfully logged into the Ubuntu Server.

4. Basic System Setup: I updated and upgraded the Ubuntu system using the following commands:

- sudo apt update
- sudo apt upgrade -y

I also installed essential packages such as curl, wget, git, nano, net-tools and unzip.

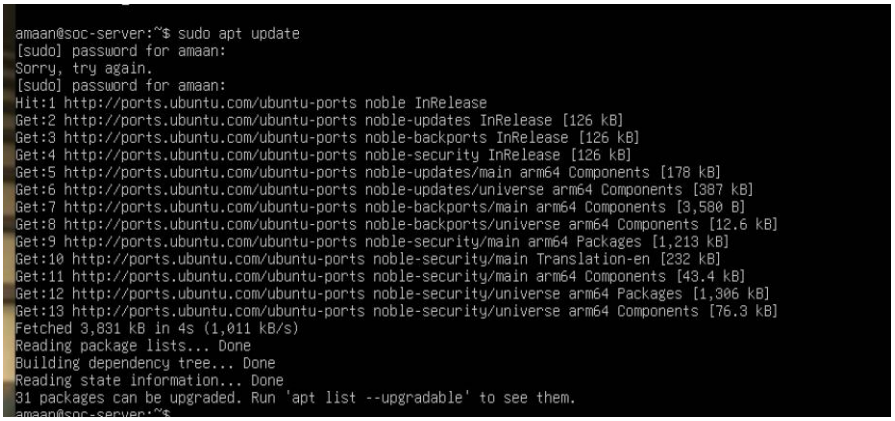


Figure 2: Successful execution of system package update using sudo apt update.

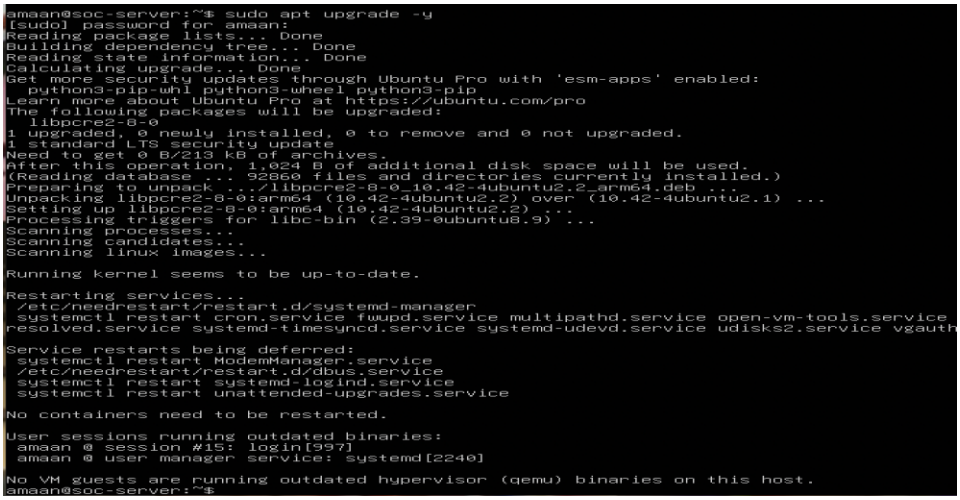


Figure 3: Successful System Upgrade on Ubuntu Server using sudo apt upgrade -y.



Figure 5: Essential Packages Installation and Verification on Ubuntu Server.

5. Hostname Configuration: I configured the server hostname as soc-server. The hostname was verified successfully using the terminal.

6. Network Configuration: I checked the network configuration and confirmed that the network interface was active and connected.

- Network Interface: enp2s0
- IP Address: 172.16.11.128
- Network Status: UP

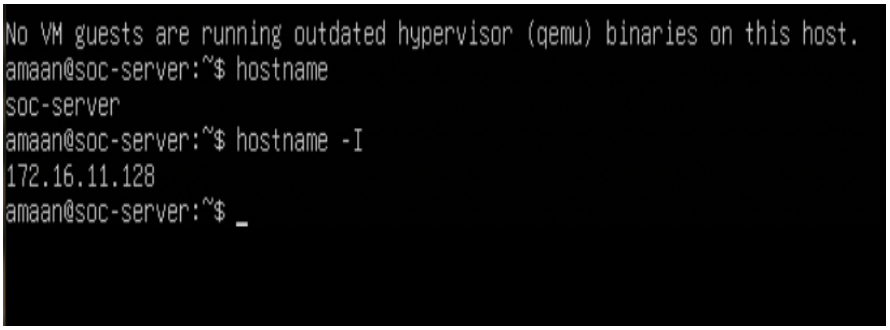


Figure 4: Hostname and IP Address Verification on Ubuntu Server.

7. Python Installation: I installed Python 3 and a pip for future scripting and automation activities.

- Python Version: 3.12.3
- pip Version: 24.0

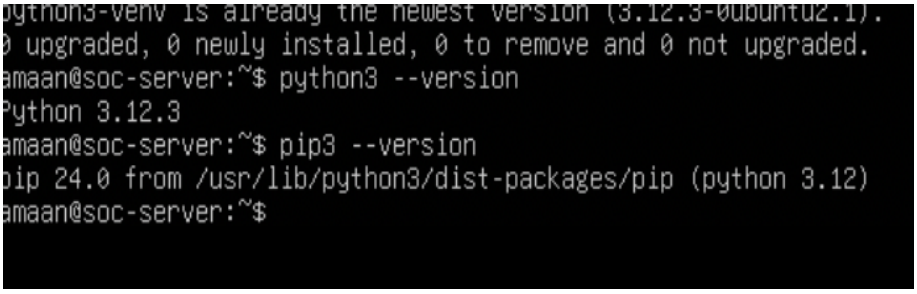


Figure 6: Python and pip Version Verification on Ubuntu Server.

8. Result: Successfully installed and configured VMware Fusion and Ubuntu Server 24.04.5 LTS. The system was updated, the hostname was configured, and network connectivity was verified. The environment is now ready for the next task, which involves installing and configuring Snort IDS.

Task 3 - Install Snort IDS

1. Objective: The objective of this task was to install Snort, an Intrusion Detection System (IDS), on Ubuntu Server and verify that it was working correctly.

2. Snort Installation: Snort was installed on Ubuntu Server 24.04.5 LTS using the following commands:

sudo apt update

sudo apt install snort –y

During installation, I configured the local network range as 172.16.11.0/24 and selected the network interface enp2s0.

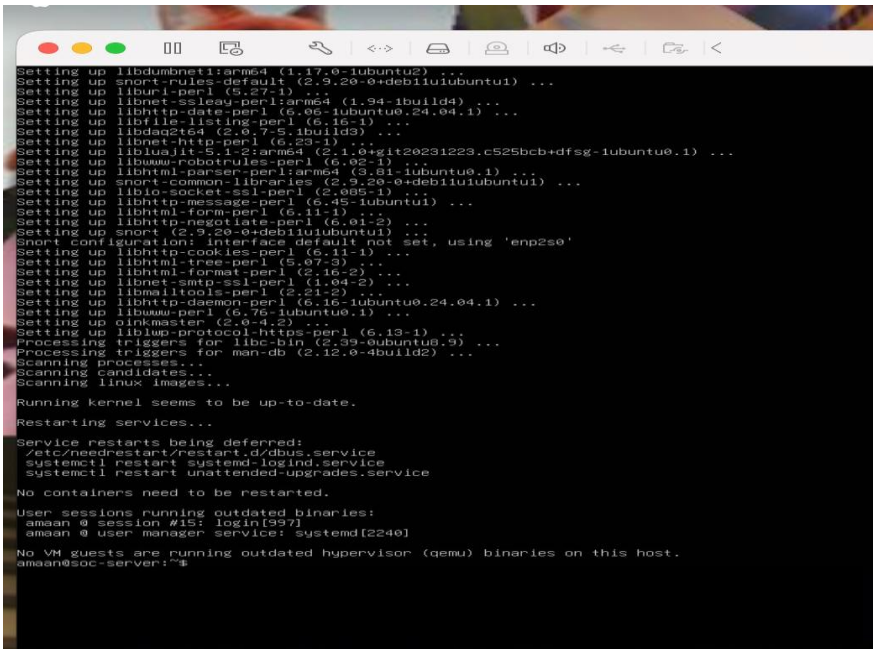


Figure 7: Snort IDS Installation on Ubuntu Server

3. Snort Version Verification: After installation, I checked the Snort version using:

snort –V

The command successfully displayed Snort Version 2.9.20 GRE (Build 82), confirming that Snort was installed correctly.

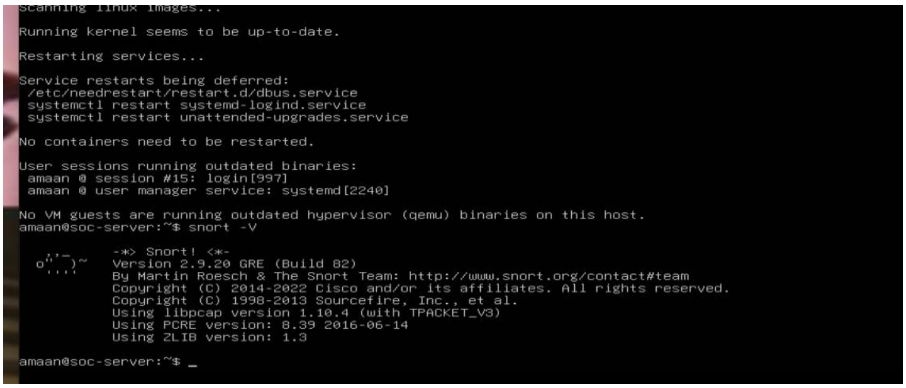


Figure 8: Snort IDS Version Verification

4. Snort Configuration Testing: I tested the Snort configuration using:

sudo snort -T -c /etc/snort/snort.conf

The test completed successfully and displayed the message:

Snort successfully validated the configuration!

This confirmed that Snort's configuration file was valid.

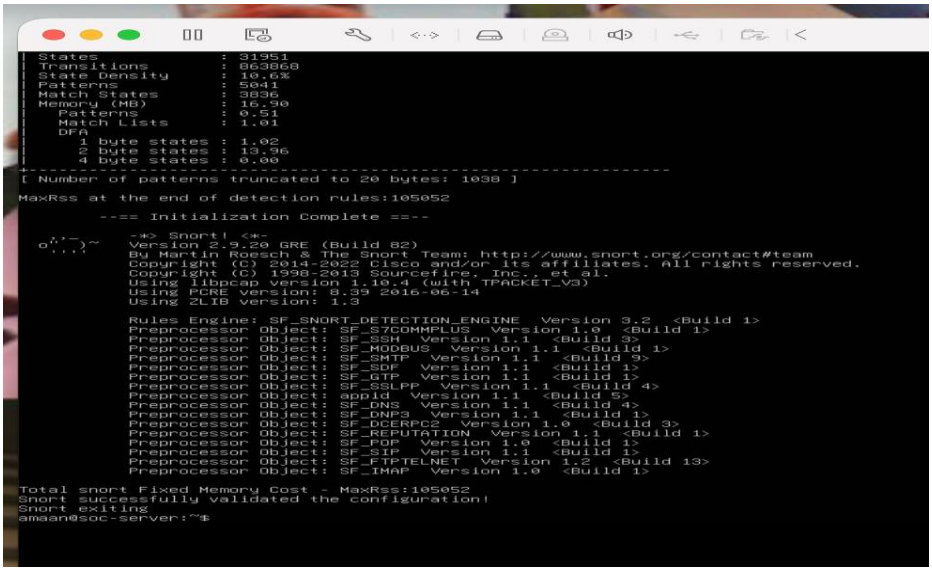


Figure 9: Snort Configuration Validation Test

5. Result: Snort IDS was successfully installed and configured on Ubuntu Server. The installation was verified, and the configuration test passed successfully. The system is now ready for further network security monitoring activities.

Task 4: Understanding and Creating Custom Snort Rules.

1. Objective: The objective of this task was to create custom detection rules in Snort and test them by generating network traffic. The purpose was to understand how Snort detects ICMP ping requests and HTTP traffic using custom rules.

2. Custom Snort Rules Configuration

I opened the local Snort rules file using the following command:

sudo nano /etc/snort/rules/local.rules

I created two custom detection rules.

Rule 1 – ICMP Ping Detection

alert icmp any any -> $HOME_NET any (msg:"ICMP Ping Detected"; sid:1000001; rev:1;)

This rule detects ICMP traffic, such as ping requests, directed towards the local network.

Rule 2 – HTTP Traffic Detection

alert tcp any any -> $HOME_NET 80 (msg:"HTTP Traffic Detected"; sid:1000002; rev:1;)

This rule detects TCP traffic directed towards HTTP port 80 on the local network.

Both rules were saved in the local.rules file.



Figure 10 – Custom Snort Rules Configuration

3. Configuration Validation

After creating the rules, I checked the Snort configuration using:

sudo snort -T -c /etc/snort/snort.conf

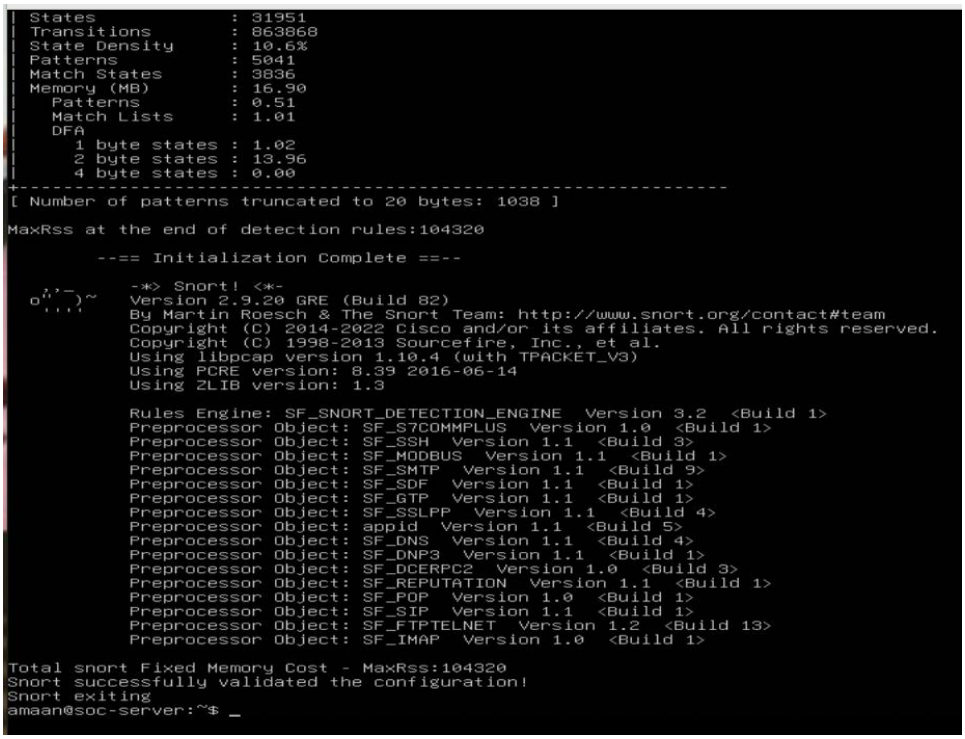


Figure 11 – Custom Snort Rules Configuration Validation

Snort successfully validated the configuration without errors.

4. ICMP Ping Detection

I started Snort in live monitoring mode using:

sudo snort -i enp2s0 -A console -c /etc/snort/snort.conf

I generated ping traffic to test the ICMP detection rule.

Snort successfully detected the traffic and displayed the alert message:

ICMP Ping Detected

This confirmed that the custom ICMP detection rule was working correctly.

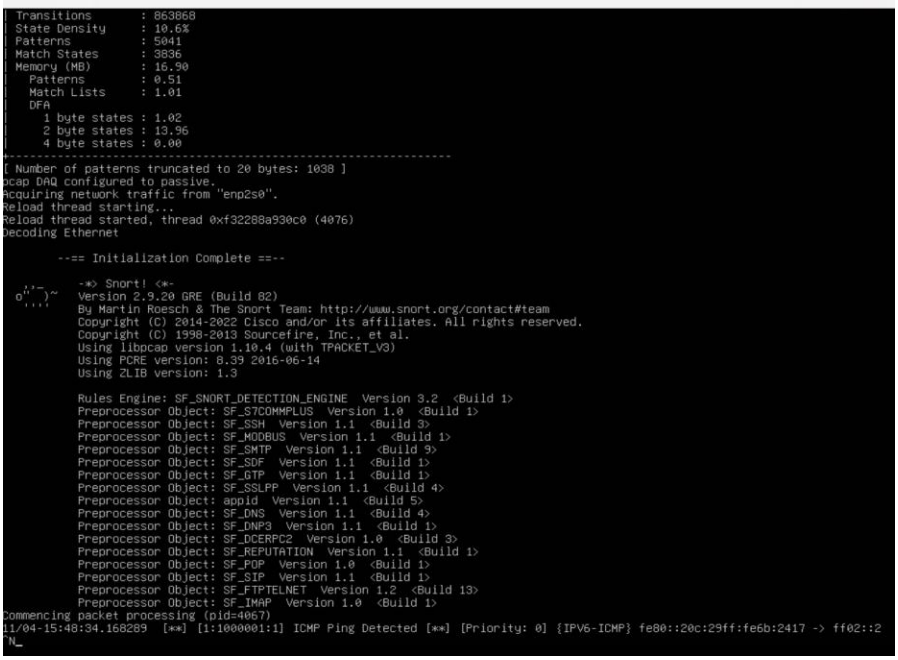


Figure 12 – ICMP Ping Detection Using Custom Snort Rule

5. HTTP Traffic Detection

To test the HTTP detection rule, I created a simple local web page and started a Python HTTP server on port 80.

I accessed the local web server from my Mac browser using the Ubuntu server's IP address.

Snort successfully detected the HTTP traffic and displayed:

HTTP Traffic Detected

The alert showed the source IP, destination IP and HTTP port 80.



Figure 13 – Successful HTTP Request to Local Ubuntu Web Server

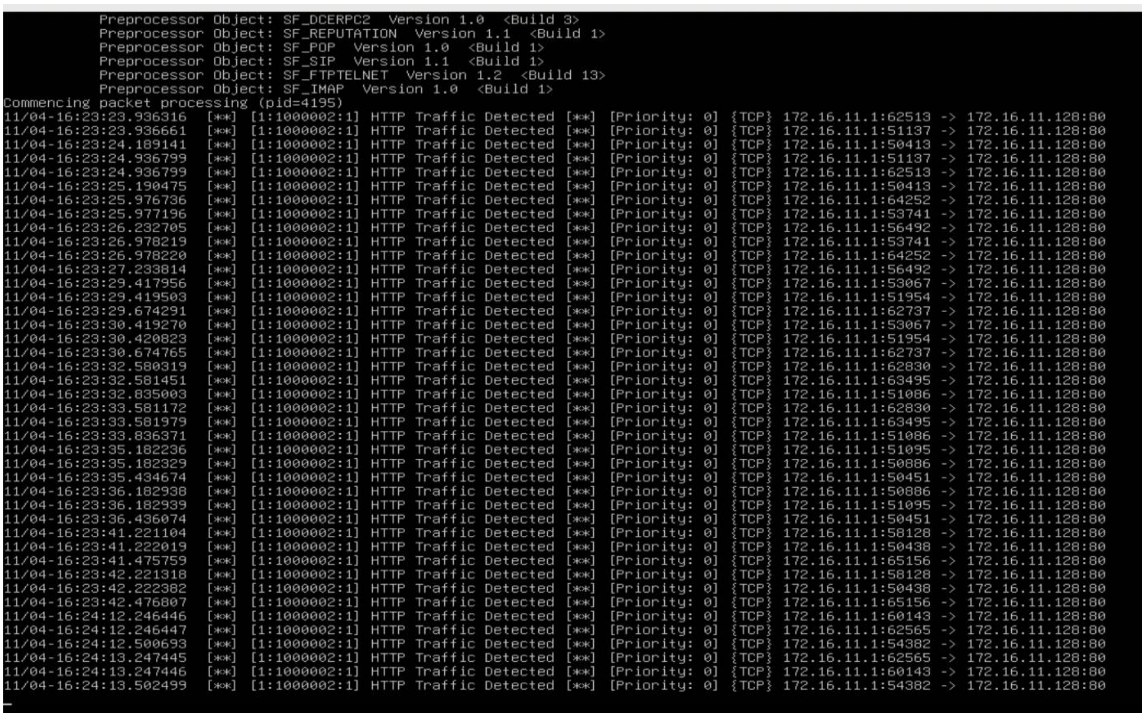


Figure 14 – HTTP Traffic Detection Using Custom Snort Rule

6. Result

Both custom Snort rules were successfully created and tested. Snort detected ICMP ping traffic and HTTP traffic as expected. This practical helped me understand how custom IDS rules can be created and used to monitor network activity.

Task 5 – Testing

1. Objective: The objective of this task was to test Snort IDS by generating network traffic and checking whether it successfully detected suspicious or monitored activities through custom rules.

2. Ping Test (ICMP Traffic)

I generated ICMP ping traffic to test the custom ICMP detection rule.

I used the following command:

ping -c 4 172.16.11.128

Snort successfully detected the ping traffic and generated the alert:

ICMP Ping Detected

This confirmed that the ICMP detection rule was working correctly.

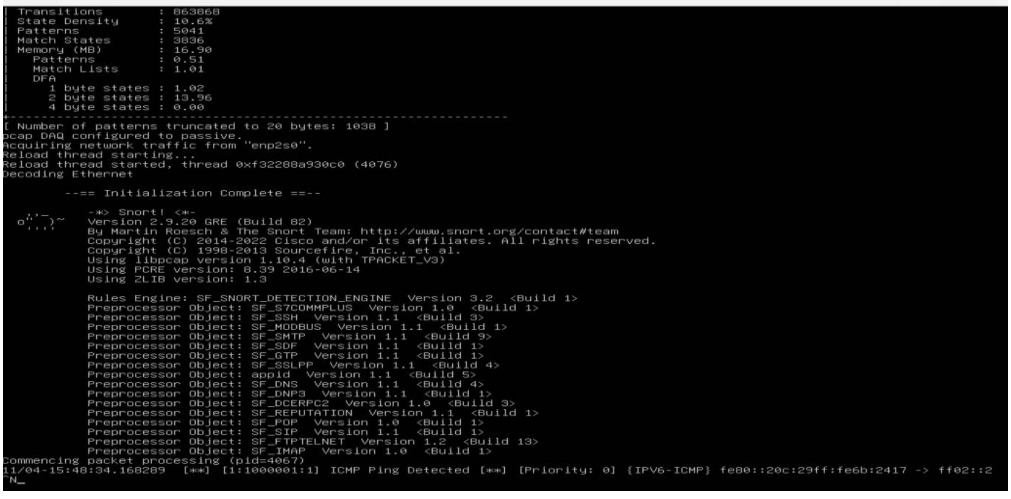


Figure 15: ICMP Ping Detection

3. Website Testing (HTTP Traffic)

I created a simple local web page and started a Python HTTP server on port 80.

I accessed the local website from my Mac browser using the Ubuntu Server IP address:

http://172.16.11.128

The web server successfully received the HTTP request.

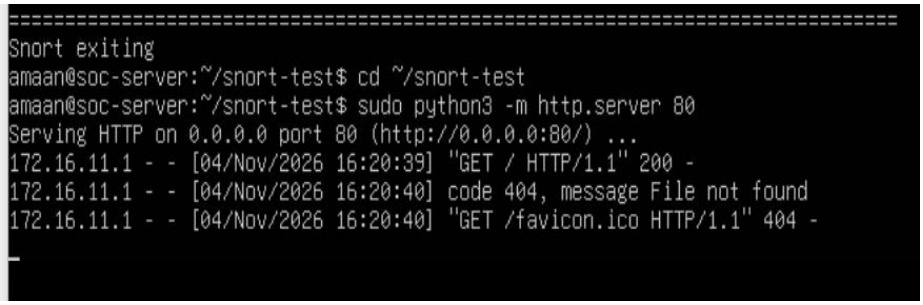


Figure 16: Successful HTTP Request to Local Ubuntu Web Server

4. Alert Detection

While monitoring network traffic, Snort successfully detected HTTP traffic and generated the alert:

HTTP Traffic Detected

The alert displayed the source IP address, destination IP address and destination port.

This confirmed that the custom HTTP detection rule was working correctly.

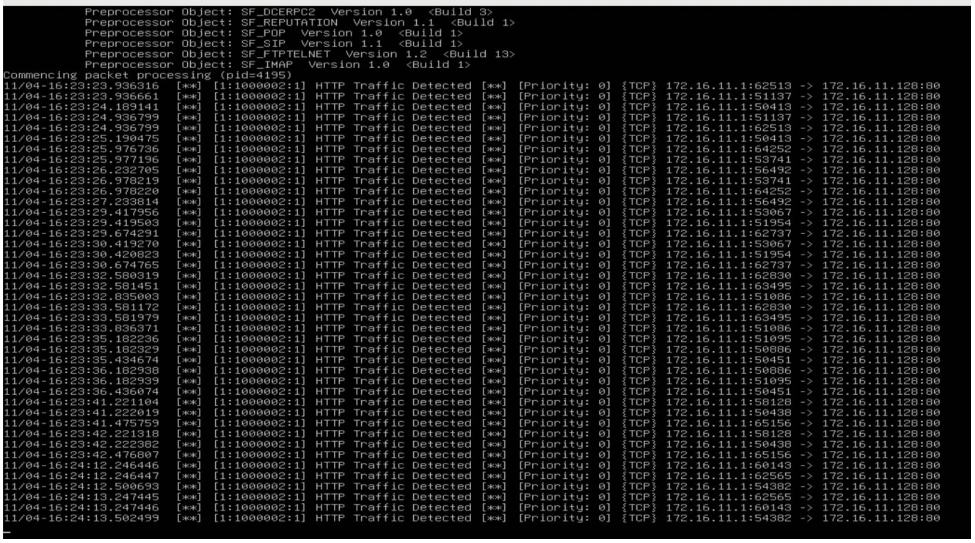


Figure 17: HTTP Traffic Detection Using Custom Snort Rule

5. Result

Both ICMP and HTTP traffic tests were successful. Snort generated the expected alerts for both types of traffic. This practical demonstrated how an Intrusion Detection System can monitor network activity and identify traffic based on custom detection rules